

Shubh Kapoor

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TECHNICAL SKILLS

AI/ML: Python, PyTorch, scikit-learn, Deep Learning, Computer Vision, NLP, Model Training Fine-tuning
LLMs Agents: LLM APIs, Agentic RAG, Tool/Function Calling, ReAct, Multi-Agent Systems, LangGraph, LangChain, LlamaIndex, Agent Memory, Context Engineering, MCP
Retrieval Knowledge Systems: RAG, Hybrid Search, GraphRAG, Embeddings, Reranking, Vector Graph Databases (ChromaDB, FAISS, Pinecone, Weaviate, Neo4j)
Agent Reliability Evaluation: Agent Evals, Benchmarking, LLM-as-Judge, Regression Testing, Agent Tracing, Observability, Failure Recovery, Durable Workflows, Human-in-the-Loop
AI Infrastructure MLOps: Docker, CI/CD, GitHub Actions, AWS Bedrock, MLflow, Model Serving, LLM Inference, Cost/Latency Optimization, LLM/Agent Observability and Monitoring
Software Engineering: FastAPI, React.js, TypeScript, JavaScript, Node.js, SQL, Git/GitHub

PROJECTS

Sentinel-47: AI Energy Supply Risk Intelligence Platform *Next.js, React, TypeScript, Gemini 2.5 Flash, Zustand, TanStack Query, SQLite, Tailwind CSS* [Live Demo](#)

- Engineered a multi-agent geopolitical intelligence platform integrating live market telemetry, corridor risk assessment, disruption simulation, and procurement optimization through four coordinated AI modules.
- Developed an LLM-powered scenario modelling engine with historical backtesting, weighted composite risk analysis, streaming executive memo generation, cryptographic audit ledger, and resilient offline fallback architecture.
- Implemented enterprise-grade frontend using Zustand state management, TanStack Query caching, deep-linkable dashboard sessions, guided demo workflows, error boundaries, and responsive visualization pipelines.

Healthcare Cost Intelligence *Python, SQLite, ChromaDB, sentence-transformers, FastAPI, Vercel* [Live Demo](#)

- Built an offline-first AI clinical decision tool taking patient inputs (symptom, city, age, comorbidities, budget) and returning explainable 5-component cost estimates with confidence scores, zero paid API dependencies.
- Engineered a rule-based intent mapper across 50+ procedures with city-tier, age, comorbidity, and ICU multipliers; applied semantic retrieval via ChromaDB and HuggingFace sentence-transformers across 43 hospital records.
- Designed a weighted multi-factor ranking algorithm covering clinical match, reputation, accessibility, and affordability; exposed results via a FastAPI REST endpoint and deployed the full stack to Vercel.

WORK EXPERIENCE

AI/ML Intern

May 2026 – Aug 2026

FintronicsAI Pvt. Ltd.

- Built a real-time driver monitoring system using YOLOv8 and MediaPipe Face Mesh; computed EAR and MAR with adaptive 3-second calibration to detect drowsiness, yawning, and distraction.
- Improved smartphone detection by curating 465 task-specific images in Roboflow from an initial 12,700-image dataset, reducing missed detections and false positives in representative driver-monitoring scenarios.
- Researched 7 financial-AI approaches spanning sentiment analysis, LSTM-DNNs, FinBERT, multimodal learning, and LLMs; proposed an adaptive multimodal framework combining news sentiment and market indicators.

UI/UX Design Intern

Dec 2025 – Jan 2026

TalentiQo Workforce Pvt. Ltd.

- Designed wireframes and high-fidelity Figma screens for 2 product features across 3 iteration cycles; created a 20-component reusable UI library covering buttons, inputs, cards, navigation, and common interaction states.
- Prepared developer handoff documentation for 2 feature flows, specifying responsive layouts, component states, spacing, typography, and key edge cases to support implementation.

EDUCATION

The LNM Institute of Information Technology

Jaipur, India

B.Tech in Computer and Communication Engineering

2023 – 2027

Dr. Virendra Swarup Education Centre

Kanpur, India

High School, Science (ISC)

2021 – 2023